



WALNUT IMPROVEMENT PROGRAM

PROJECT LEADER: Pat J. Brown, Department of Plant Sciences, UC Davis

COLLABORATORS: Chuck Leslie, Kristina McCreery, Laia Menendez Diaz, Zach Uebelhor, Steven Lee, Franklin Lewis, Wes Hackett, Ivan Bermudez, Matt Davis, and Jeffrey Yen (Department of Plant Sciences, UC Davis)

Project status in 2024: Ongoing with annual review

PROJECT OBJECTIVES:

1. Provide the California walnut industry with improved cultivars and rootstocks
2. Develop, maintain, and distribute germplasm with an array of useful traits for breeding
3. Apply new phenotypic, genomic and molecular tools for more efficient walnut breeding

BACKGROUND

The Walnut Improvement Program develops improved scion cultivars and new rootstocks with pathogen and abiotic stress resistance for the California walnut industry, improves our understanding of the genetics of the crop, and maintains breeding resources. Scion breeding priorities include improved blight resistance, kernel color, and kernel yield; the rootstock breeding priority is nematode resistance.

KEY FINDINGS

- Wolfskill, the latest WIP release, maintained a higher average relative value than any other cultivar in independent kernel evaluations performed by Diamond over the last 7 years, with much higher kernel yield than Chandler and similar color.
- An investigation into the causes of declining kernel yield in Chandler suggested an association with kernel weight but not shell weight; nut size also has some effect.
- A new 5-acre selection block was established at Sierra Gold with 195 different scions on RX1.
- 2078 clonal plants were rooted and acclimated; 759 were distributed to pathologists and growers.
- 17 RX1-derived rootstocks with tolerance to cherry leaf roll virus were identified and will be screened for resistance to Phytophthora and salinity.
- Machine vision methods were deployed to estimate kernel color and size and are being

developed to count nuts and estimate yield from image data.

- Biotechnology tools are being developed to enable speed breeding and transgene-free genome editing.